# BUSFIN 8200: Problem Set 2

Xi Wang

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## Question 1

### (a)

We have complete markets iff we can find a unique matrix $\Omega$ s.t. $X\Omega = I$, where

$$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}, \qquad \Omega = \begin{bmatrix} | & | & | \\ w_1 & w_2 & w_3 \\ | & | & | \end{bmatrix},$$

and each $w_i$ is a portfolio weight vector using A, B and C. For example, $(10, 20, 30)'$ is asset A's payoff across the three states.

$X\Omega = I$ means

$$Xw_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad Xw_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \qquad Xw_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Because $X$ is $3 \times 3$, $X\Omega = I$ has a unique solution iff $X$ is invertible, and

$$X \text{ invertible} \iff \det(X) \neq 0 \iff \operatorname{rank}(X) = 3.$$

$$X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix} = 10 \begin{bmatrix} 1 & 1 & 0 \\ 2 & 1 & 1 \\ 3 & 1 & 1 \end{bmatrix}.$$

Row reduction:

$$\begin{aligned} \text{row } 3 - \text{row } 2: &\quad \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{exchange rows 1 and 2}: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{row } 2 - \text{row } 1: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 1 & 1 \end{bmatrix} \\ \text{row } 3 - 3 \cdot \text{row } 1 - \text{row } 2: &\quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Thus $\operatorname{rank}(X) = 3$ and $X$ is invertible. $\Omega = X^{-1}$ exists and is unique, with $X\Omega = XX^{-1} = I$. Thus, the market is complete.

## (b)

Let $X$ be the matrix of $N = 3$ non-redundant asset payoffs, let $P$ be the price vector of the original assets, and let $P_c = (q_1, q_2, q_3)'$ be the price vector of the contingent claims. Use the LOP to solve for $P_c$ as a function of $X$ and $P$:

$$P = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}, \qquad X = \begin{bmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{bmatrix}.$$

The LOP says that two assets with identical payoffs in every state of nature must have the same price. Therefore,

$$X'P_c = P \quad \Longrightarrow \quad \begin{bmatrix} 10 & 20 & 30 \\ 10 & 10 & 10 \\ 0 & 10 & 10 \end{bmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \begin{pmatrix} 16 \\ 8 \\ 6 \end{pmatrix}.$$

Dividing everything by 10:

$$\begin{cases} q_1 + 2q_2 + 3q_3 = 1.6 \\ q_1 + q_2 + q_3 = 0.8 \\ q_2 + q_3 = 0.6 \end{cases} \quad \Longrightarrow \quad \begin{cases} q_1 = 0.2 \\ q_2 = 0.4 \\ q_3 = 0.2 \end{cases}$$

Thus, $P_c = (0.2, 0.4, 0.2)'$.

## (c)

$$P_D = \text{Payoff}(D) \cdot P_c = \begin{pmatrix} 30 & 20 & 10 \end{pmatrix} \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix} = 6 + 8 + 2 = 16.$$

## (d)

$$P(X) = E[MX] = \sum_{s=1}^{S} \pi(s) M(s) X(s).$$

Since $q_s = \pi(s)M(s)$, we have $M(s) = q_s/\pi(s)$. With

$$P_c = \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix}, \qquad \pi(s) = \begin{pmatrix} 1/4 \\ 1/2 \\ 1/4 \end{pmatrix} \quad \Longrightarrow \quad M(s) = \begin{pmatrix} 0.8 \\ 0.8 \\ 0.8 \end{pmatrix}.$$

By definition, no arbitrage means that

1. An asset with payoff $\geq 0$ must have price $\geq 0$.
2. An asset with payoff $\geq 0$ and a chance for payoff $> 0$ must have price $> 0$.

We have $X \geq 0$, $\Pr(X > 0) > 0$, and $P(X) = E[MX] > 0$ (since $M > 0$). Thus, there is no arbitrage.

Moreover, $P(X) = E[MX] = 0.8\,E[X]$. The SDF stays constant across the three states, which implies it is risk-neutral.

**(e)**

$$\begin{aligned} P_D = E[MX_D] &= \sum_{s=1}^{S} \pi(s) M(s) X_D(s) \\ &= \tfrac{1}{4} \cdot 0.8 \cdot 30 + \tfrac{1}{2} \cdot 0.8 \cdot 20 + \tfrac{1}{4} \cdot 0.8 \cdot 10 \\ &= 6 + 8 + 2 = 16. \end{aligned}$$

This is the same as in Q1(c).

# Question 2

## (a)

I construct the unconditional linear SDF as

$$\hat{M}_t^* = \hat{a} - \hat{b}' x R_t,$$

where $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}$ and $\hat{a} = 1 + \hat{b}'\hat{\mu}$, imposing the normalization $E[M^*] = 1$.

Table 1 reports the estimated SDF coefficients. The estimated SDF satisfies the unconditional pricing conditions, with $\widehat{E}[\hat{M}^*] = 1$ and $\widehat{E}[\hat{M}^* x R_j] \approx 0$ for all 10 portfolios.

Figure 1 shows no clear monotonic relationship between the estimated $b_j$ coefficients and book-to-market deciles. Instead, the coefficients fluctuate considerably across portfolios, with both positive and negative values.

This pattern arises because the coefficients are determined by both the expected excess returns and the covariance structure of the portfolios, as reflected in $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}$. Therefore, portfolios with higher average excess returns do not necessarily receive larger positive coefficients.

Table 1: Estimated BM SDF intercept and coefficients, rounded to four decimal places. Each $\hat{b}_j$ corresponds to BM decile $j$; estimation uses decimal monthly excess returns.

| $\hat{a}$ | $\hat{b}_1$ | $\hat{b}_2$ | $\hat{b}_3$ | $\hat{b}_4$ | $\hat{b}_5$ | $\hat{b}_6$ | $\hat{b}_7$ | $\hat{b}_8$ | $\hat{b}_9$ | $\hat{b}_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1.0368 | -1.5850 | 3.5503 | 0.7104 | -3.1122 | 1.1837 | 2.9972 | -5.2931 | 2.4722 | 2.8775 | -1.1771 |

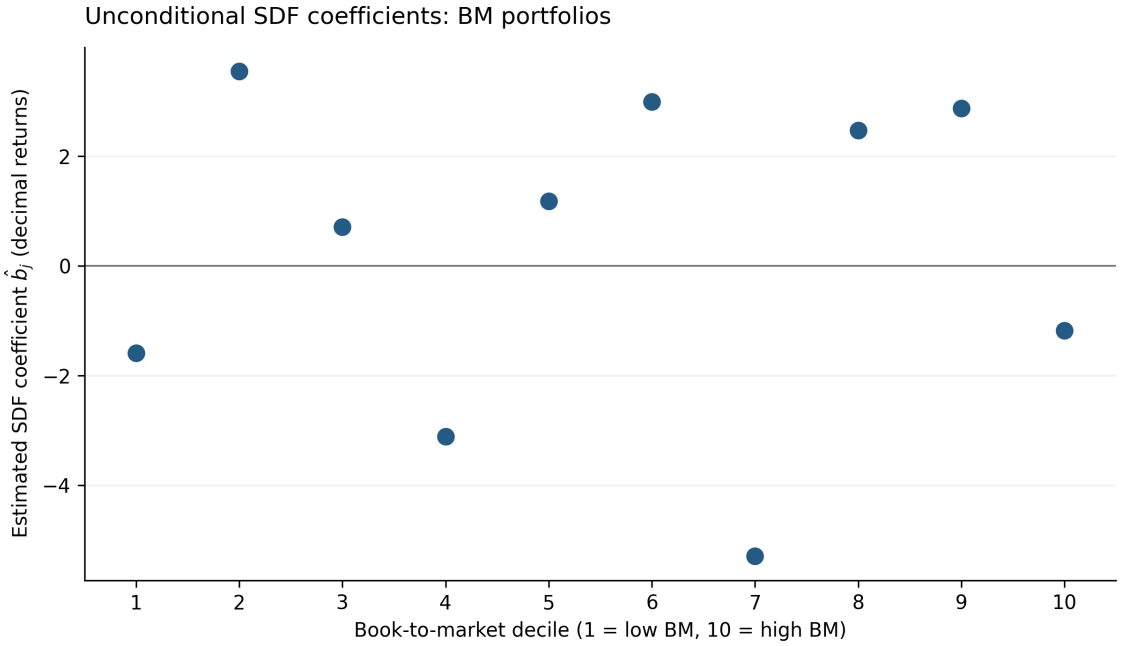



Figure 1: Estimated coefficients $\hat{b}_j$ in the unconditional SDF $\hat{M}_t^* = \hat{a} - \hat{b}' x R_t$, normalized so that $\widehat{E}[\hat{M}^*] = 1$. The sample contains 1,190 overlapping monthly observations from July 1927 through August 2026, as specified in the analysis prompt. Returns are in decimals, and the sample covariance matrix uses divisor $T$.

## (b)

Following the same approach as in part (a), I construct the unconditional linear SDF using the 10 momentum portfolios. Table 2 reports the estimated SDF coefficients.

Figure 2 shows that the estimated coefficients vary considerably across momentum deciles, with both positive and negative values. There is no clear monotonic pattern. As in part (a), this pattern reflects both the expected excess returns and the covariance structure of the portfolios.

The sample correlation between the BM and MOM SDFs is 0.3555, indicating a moderate positive relationship. This suggests that the two SDFs capture some common variation, but also reflect different components of asset returns. The sample correlation between the BM and MOM SDFs is

$$\widehat{\mathrm{Corr}}\left(\hat{M}^{*,BM}, \hat{M}^{*,MOM}\right) = 0.3555.$$

This uses 1,190 aligned monthly observations from July 1927 through August 2026.

Table 2: Estimated MOM SDF intercept and coefficients, rounded to four decimal places. Each $\hat{b}_j$ corresponds to MOM decile $j$; estimation uses decimal monthly excess returns.

| $\hat{a}$ | $\hat{b}_1$ | $\hat{b}_2$ | $\hat{b}_3$ | $\hat{b}_4$ | $\hat{b}_5$ | $\hat{b}_6$ | $\hat{b}_7$ | $\hat{b}_8$ | $\hat{b}_9$ | $\hat{b}_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1.0666 | -4.0194 | 1.1429 | 0.1024 | 1.4897 | 2.5317 | -0.8195 | 0.2777 | 0.9818 | -0.9414 | 3.4370 |

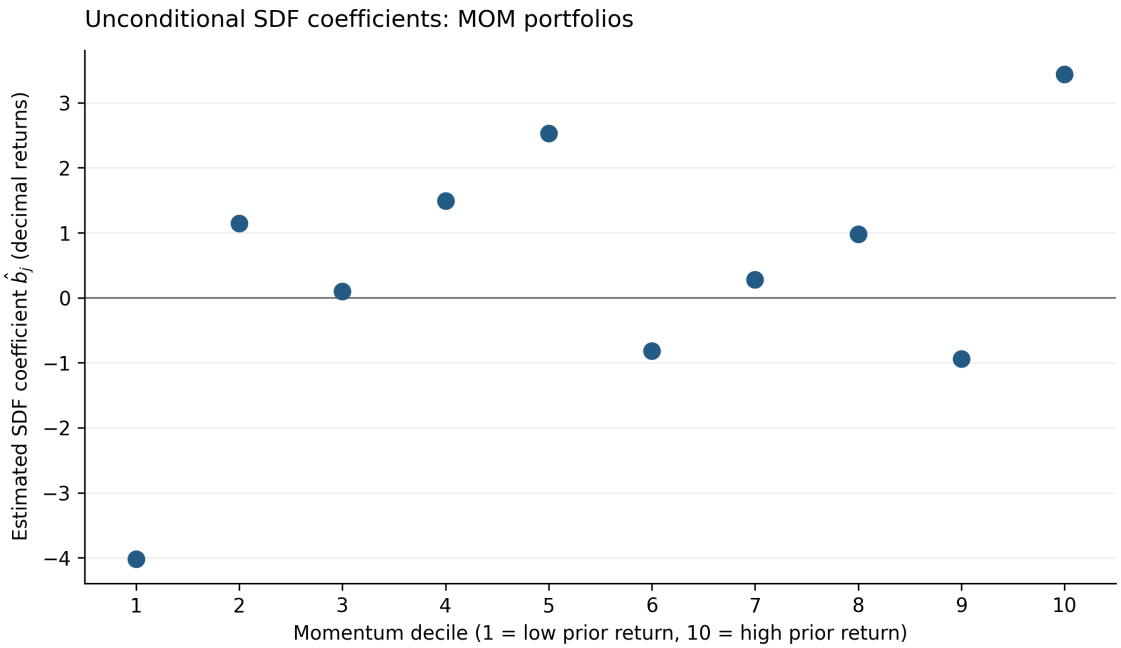



Figure 2: Estimated coefficients $\hat{b}_j$ in the unconditional SDF $\hat{M}_t^* = \hat{a} - \hat{b}' x R_t$ for the ten momentum portfolios, normalized so that $\widehat{E}[\hat{M}^*] = 1$. The sample contains 1,190 overlapping monthly observations from July 1927 through August 2026, as specified in the analysis prompt. Returns are in decimals, and the sample covariance matrix uses divisor $T$, as in part (a).